Online Material: Scale over Engagement

O1 Matching Overlap and Omitted-Confounding Sensitivity

O1.1 Propensity-Score Overlap

We report propensity-score overlap before and after matching to evaluate common support and verify covariate balance, respectively [2, 3]. Pre-match distributions assess whether treated and control units share the same covariate space; a lack of common support would restrict the matched contrast to a smaller population. Post-match scores confirm whether matching substantially aligned these distributional differences, with residual separation indicating remaining divergence. For each task, we fitted a logistic regression on the pre-match pool using the continuous matching covariates, and applied that model unchanged to the matched sample.

As shown in Fig. OF1, the overlapping coefficient (the shared area of the density histograms on $[0, 1]$) improved substantially across all tasks after matching: from 0.888 to 0.978 for creation, 0.488 to 0.846 for consumption, 0.790 to 0.991 for algorithmic distribution, and 0.978 to 0.999 for algorithm design. Pre-match common support was already strong; only the consumption task had a notable number of treated units (2,618) falling outside the control score range, whereas the other three tasks had negligible out-of-support units (0, 1, and 0, respectively). Consumption also exhibited the most distinct pre-match density shape, but successfully aligned after matching with both densities peaking near 0.6. For creation and algorithm design, scores remained near zero throughout, reflecting their low treated shares (2.8% and 0.99%) in the candidate pools.

O1.2 Sensitivity Analysis

O1.2.1 Methodological Framework. While our matching procedure ensures strict comparability across observed characteristics, observational estimates remain inherently susceptible to unobserved confounding. To systematically assess the robustness of our estimates against potential omitted variable bias, we implemented the sensitivity analysis framework introduced by Cinelli and Hazlett [1]. The core intuition of this framework is: rather than relying on the untestable assumption that no unobserved confounders exist, it quantifies exactly how influential such an unobserved confounder must be to overturn the results.

This quantification is achieved by reparameterizing the required strength of unobserved confounding in terms of partial variance explained (R^2). Specifically, the baseline strength of the estimated impact is first captured by $R^2_{Y\sim D|\mathbf{X}}$, representing the proportion of residual variance in the outcome uniquely explained by the treatment assignment (D) after controlling for all observed covariates (X). Building upon this baseline strength, the framework calculates the Robustness Value (RV). $RV_{q=1}$ defines the critical threshold: the minimum explanatory power a hypothetical unobserved confounder (Z) must share with both the treatment and the outcome to completely eliminate the estimated impact. Correspondingly, $RV_{q=1,\alpha=0.05}$ defines the threshold required to render the point estimate statistically insignificant at the 5% level, which serves as a more conservative robustness bound than $RV_{q=1}$.

To assess whether an unobserved confounder could realistically reach the RV threshold, this framework benchmarks it against the empirical strength of the observed covariates. Specifically, it calculates how strongly each known variable (Z_k) influences the treatment assignment ($R^2_{D\sim Z_k|\mathbf{X}}$) and the outcome ($R^2_{Y\sim Z_k|\mathbf{X},D}$). Because our models already incorporated highly predictive factors as confounders—such as historical engagement and creator influence—the maximum explanatory power among these known variables serves as a conservative empirical bound. If the calculated

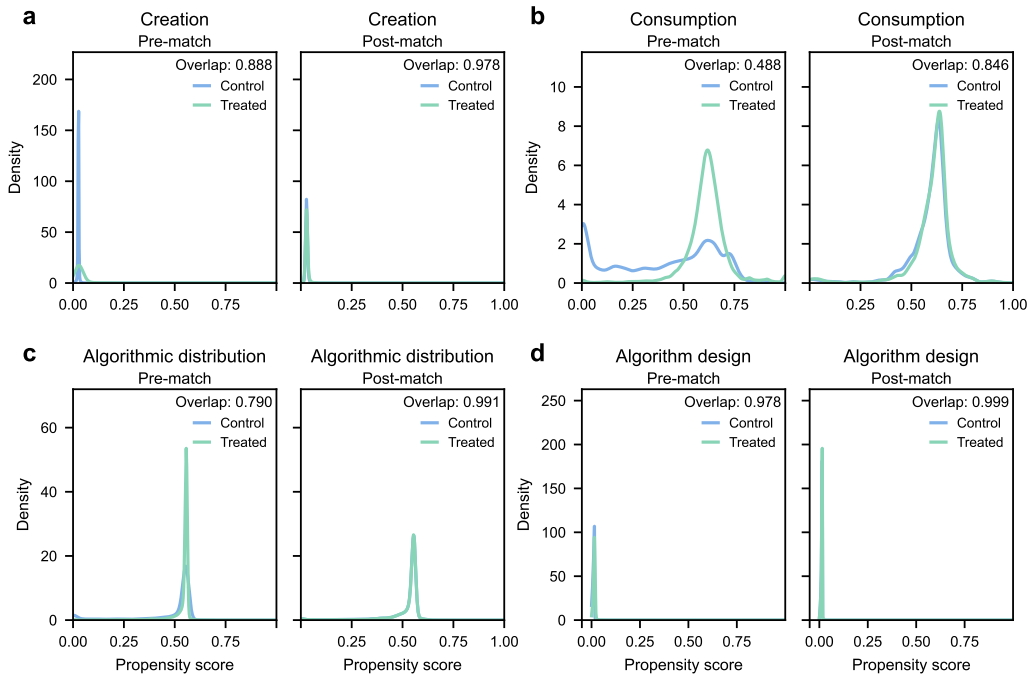


Fig. OF1. **Propensity-score distributions before and after matching. a**, AIGC-adopting and non-adopting creators; **b**, interactions with AIGC and non-AIGC videos; **c**, AIGC and non-AIGC videos in the exposure comparison; **d**, consumers receiving population-feedback and individual-feedback content distribution algorithms. Within each task, the left panel shows the candidate pool and the right panel the matched sample. Scores were estimated using a logistic model fitted to the pre-match pool with the continuous matching covariates and applied unchanged to the matched sample. Green denotes the focal group (AIGC-adopting creators, interactions with AIGC, AIGC videos, or consumers receiving the population-feedback content distribution algorithm); blue denotes its comparison group. Overlap is the shared area of density histograms using 50 equal-width bins on $[0, 1]$.

RV substantially exceeds this bound, the results are deemed highly robust. Consequently, if the calculated *RV* is multiple times larger than these bounds ($R^2_{D\sim Z_k|\mathbf{X}}$ and $R^2_{Y\sim Z_k|\mathbf{X},D}$), overturning the results would require an unobserved confounder vastly more powerful than the observed primary drivers, supporting the robustness of our estimates relative to observed baseline covariates.

O1.2.2 Interpretation of Results. We applied this sensitivity framework across the four primary analytical dimensions. The results consistently validated the robustness of our core findings, quantifying the substantial confounding strength required to overturn both the behavioral shifts and the structural exposure patterns.

- **Creation-side analysis (Table OT1):** The sensitivity analysis differentiates the robust impact on productivity from the marginal differences in aggregate returns. The estimated increase in Video Creation Volume is highly robust ($RV_{q=1} = 0.2635$); an unobserved confounder would need to exert an influence more than five times greater than our maximal observed benchmark ($R^2_{D\sim Z|\mathbf{X}} = 0.0488$, $R^2_{Y\sim Z|\mathbf{X},D} = 0.0224$) to eliminate this effect. Conversely, the minimal R^2 and RV values for Total Valid Views ($R^2_{Y\sim D|X} = 0.0004$, $RV_{q=1} = 0.0198$) and

Total Full Views ($R^2_{Y\sim D|X} = 0.0013$, $RV_{q=1} = 0.0355$) reflect the near-zero baseline differences between groups, consistent with statistical equivalence in aggregate engagement returns. This lack of a robust divergence in aggregate returns naturally aligns with our baseline observation: while AIGC adoption systematically scaled up production volume, aggregate engagement returns were comparable to those of non-adopting creators.

Table OT1. **Sensitivity analysis for creation-side behavioral outcomes to omitted confounding.** *Coef.* reports the regression coefficient of the treatment (creator's adoption of the official tool) for each respective outcome (creator behavioral metrics). $R^2_{Y\sim D|\mathbf{X}}$ denotes the proportion of residual outcome variance explained by the treatment. $RV_{q=1}$ and $RV_{q=1,\alpha=0.05}$ represent the robustness values required for an unobserved confounder to reduce the estimate to zero or render it statistically insignificant at the 5% level, respectively. The reference columns report the maximum explanatory power of the observed covariates (Z_k) on the treatment ($R^2_{D\sim Z_k|\mathbf{X}}$) and the outcome ($R^2_{Y\sim Z_k|\mathbf{X},D}$), identified by iterating through all candidates to establish an empirical benchmark. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Outcome	*Coef.*	$R^2_{Y\sim D\|\mathbf{X}}$	$RV_{q=1}$	$RV_{q=1,\alpha}$	$R^2_{D\sim Z_k\|\mathbf{X}}$	$R^2_{Y\sim Z_k\|\mathbf{X},D}$
Video creation volume (log1p)	0.5548^{***}	0.0862	0.2635	0.2425	0.0488	0.0224
Total valid views (log1p)	0.0836	0.0004	0.0198	0.0000	0.0488	0.0105
Total full views (log1p)	-0.1484^{*}	0.0013	0.0355	0.0077	0.0488	0.0114

- **Consumption-side analysis (Table OT2):** The estimated engagement disparity between AIGC and non-AIGC, conditional on algorithmic exposure, is robust to unobserved confounding.
  For the valid-view rate, the treatment coefficient is −0.0507 and the robustness value is $RV_{q=1} = 0.0923$. This threshold is about 30 times the explanatory power of the strongest observed covariate ($R^2_{D\sim Z|\mathbf{X}} = 0.0030$). The same pattern holds for the full-view rate, the early-retention rate, view duration, and the normalized view fraction: all five coefficients are negative and statistically significant, and each $RV_{q=1}$ exceeds that benchmark by a factor of about 14 to 33. View duration has the smallest robustness value ($RV_{q=1} = 0.0407$), which is still above 0.0030. Including video duration as a covariate leaves these estimates nearly unchanged; the normalized view fraction moves slightly further from zero (from −0.0478 to −0.0495). The view-duration coefficient is a mean difference of −0.99 s. A Hodges–Lehmann estimate for the same outcome is −0.115 s; the two quantities are not interchangeable.
- **Algorithmic distribution analysis (Table OT3):** The estimated reduction in the algorithmic exposure of AIGC is robust to potential confounding.
  The $RV_{q=1}$ thresholds for 31-day Cumulative Exposure and Days to Reach 90% Exposure are 0.3224 and 0.3344, respectively. These values are substantially larger than the explanatory power of the primary observed pre-upload characteristics ($R^2 \approx 0.0005$). This margin indicates that the lower algorithmic exposure of AIGC is well-supported, as an unobserved confounder would need a notably larger impact than the known variables to invalidate the results.
- **Algorithm design analysis (Table OT4):** In assessing differences across algorithmic designs, the $RV_{q=1}$ for the share of total exposures allocated to AIGC (proportion scale) is estimated at 0.0948.

Table OT2. **Sensitivity analysis for consumption-side engagement outcomes to omitted confounding.** *Coef.* reports the estimated regression coefficient of the treatment (the viewed video being AIGC) for each respective outcome (consumer engagement depth), with video duration included among the covariates. $R^2_{Y\sim D|\mathbf{X}}$ denotes the proportion of residual outcome variance explained by the treatment. $RV_{q=1}$ and $RV_{q=1,\alpha=0.05}$ represent the robustness values required for an unobserved confounder to reduce the estimate to zero or render it statistically insignificant at the 5% level, respectively. The reference columns report the maximum explanatory power of the observed covariates (Z_k) on the treatment ($R^2_{D\sim Z_k|\mathbf{X}}$) and the outcome ($R^2_{Y\sim Z_k|\mathbf{X},D}$), identified by iterating through all candidates to establish an empirical benchmark. The view-duration coefficient is a mean difference in seconds. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Outcome	*Coef.*	$R^2_{Y\sim D\|\mathbf{X}}$	$RV_{q=1}$	$RV_{q=1,\alpha}$	$R^2_{D\sim Z_k\|\mathbf{X}}$	$R^2_{Y\sim Z_k\|\mathbf{X},D}$
Valid-view rate	-0.0507^{***}	0.0093	0.0923	0.0818	0.0030	0.0009
Full-view rate	-0.0213^{***}	0.0046	0.0659	0.0550	0.0030	0.0007
Early-retention rate	-0.0751^{***}	0.0109	0.0994	0.0889	0.0030	0.0013
View duration (s)	-0.9909^{***}	0.0017	0.0407	0.0296	0.0030	0.0014
Normalized view fraction	-0.0495^{***}	0.0019	0.0422	0.0310	0.0030	0.0013

Table OT3. **Sensitivity analysis for algorithmic distribution outcomes to omitted confounding.** *Coef.* reports the regression coefficient of the treatment (the video being AIGC) for each respective outcome (algorithmic exposure metrics). $R^2_{Y\sim D|\mathbf{X}}$ denotes the proportion of residual outcome variance explained by the treatment. $RV_{q=1}$ and $RV_{q=1,\alpha=0.05}$ represent the robustness values required for an unobserved confounder to reduce the estimate to zero or render it statistically insignificant at the 5% level, respectively. The reference columns report the maximum explanatory power of the observed covariates (Z_k) on the treatment ($R^2_{D\sim Z_k|\mathbf{X}}$) and the outcome ($R^2_{Y\sim Z_k|\mathbf{X},D}$), identified by iterating through all candidates to establish an empirical benchmark. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Outcome	*Coef.*	$R^2_{Y\sim D\|\mathbf{X}}$	$RV_{q=1}$	$RV_{q=1,\alpha}$	$R^2_{D\sim Z_k\|\mathbf{X}}$	$R^2_{Y\sim Z_k\|\mathbf{X},D}$
31-day exposure (log1p)	-1.7910^{***}	0.1330	0.3224	0.3202	0.0005	0.0021
Days to 90% of 31-day exposure	-5.3077^{***}	0.1438	0.3344	0.3322	0.0005	0.0005

By comparison, the strongest empirical benchmark from consumer historical profiles yields partial R^2 values of 0.0210 ($R^2_{D\sim Z|\mathbf{X}}$) and 0.0050 ($R^2_{Y\sim Z|\mathbf{X},D}$). Because the theoretical threshold for an unobserved confounder exceeds the strongest observed benchmark by more than a factor of four, the lower AIGC visibility under population feedback-driven algorithms is maintained.

Table OT4. **Sensitivity analysis for algorithm design comparison to omitted confounding.** *Coef.* reports the regression coefficient of the treatment (distribution algorithm type: 1 = individual feedback-driven, 0 = population feedback-driven) for the outcome (the share of total exposures allocated to AIGC, measured on the proportion scale). $R^2_{Y\sim D|\mathbf{X}}$ denotes the proportion of residual outcome variance explained by the treatment. $RV_{q=1}$ and $RV_{q=1,\alpha=0.05}$ represent the robustness values required for an unobserved confounder to reduce the estimate to zero or render it statistically insignificant at the 5% level, respectively. The reference columns report the maximum explanatory power of the observed covariates (Z_k) on the treatment ($R^2_{D\sim Z_k|\mathbf{X}}$) and the outcome ($R^2_{Y\sim Z_k|\mathbf{X},D}$), identified by iterating through all candidates to establish an empirical benchmark. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Outcome	*Coef.*	$R^2_{Y\sim D\|\mathbf{X}}$	$RV_{q=1}$	$RV_{q=1,\alpha}$	$R^2_{D\sim Z_k\|\mathbf{X}}$	$R^2_{Y\sim Z_k\|\mathbf{X},D}$
AIGC exposure share	0.0202^{***}	0.0098	0.0948	0.0756	0.0210	0.0050

O2 Alternative Time-series Specifications

O2.1 Granger lag specifications

Table OT5. **Granger causality tests between daily upload counts and first-day exposure.** Panel A reports AIGC and Panel B reports non-AIGC. Exposure is the median first-day exposure count for videos uploaded on each day. Tests use first-differenced daily series with lag orders from 1 to 6 days; arrows indicate the tested predictive direction. F tests the joint null that lagged values of the predictor add no forecasting information conditional on the specified model. Reported p-values use a moving-block bootstrap with 7-day blocks and 10,000 replications. Supply predicts AIGC exposure at lag orders 3–6, but not 1–2; reverse-direction tests and both directions for non-AIGC do not reject at 5%. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$; ns, not significant.

Lag	Scale → Exposure (1d)			Exposure (1d) → Scale		
	F-Stat	P-Value	Sig.	F-Stat	P-Value	Sig.
Panel A: AIGC						
1	0.537	0.467	ns	0.083	0.772	ns
2	1.130	0.329	ns	0.239	0.765	ns
3	5.541	0.012	*	0.360	0.733	ns
4	5.379	0.009	**	0.428	0.723	ns
5	4.844	0.009	**	0.407	0.774	ns
6	4.703	0.008	**	0.581	0.657	ns
Panel B: non-AIGC						
1	0.091	0.763	ns	0.016	0.897	ns
2	1.754	0.182	ns	0.038	0.961	ns
3	1.205	0.316	ns	0.120	0.945	ns
4	0.921	0.449	ns	0.096	0.984	ns
5	1.573	0.182	ns	0.124	0.985	ns
6	1.496	0.188	ns	0.202	0.973	ns

Sensitivity to alternative lag orders. To verify that our findings do not hinge on BIC lag selection, we re-estimated all six ARDL equations, allowing the lag length for each variable to vary separately up to five days. Across the examined lag specifications, the three lower-exposure-tier equations met both bounds criteria, except that the ≤10 and 101–1,000 equations yielded inconclusive results when all three lag orders are increased to five. Both engagement-depth equations cleared the bounds criteria across these specifications. Table OT6 therefore focuses on the sole borderline baseline case: the top tier (>1,000 exposures), which sat on the critical threshold ($F = 4.81$) at baseline. Because the t-statistic cleared −3.53 across all lag orders, the bounds test for a long-run relationship hinges entirely on F. Three distinct patterns emerged:

(1) **Dip and recovery with longer outcome (p) and non-AIGC-supply (q_2) lags.** Adding a second lag lowered F, whereas longer specifications restored it above the upper bound. Additional lags redistribute explanatory power between short-run dynamics and lagged-level terms, which can weaken the bounds statistic before a longer specification captures the adjustment more fully. The decline was greater for non-AIGC supply (q_2), consistent with its smooth, autocorrelated trajectory and the limited additional information in adjacent lags: F fell to 3.35 at $(1, 1, 2)$, compared with 4.48 at $(2, 1, 1)$. The upper bound was exceeded at three outcome lags (p) and five non-AIGC-supply lags (q_2).

(2) **Monotonic increase with longer AIGC-supply lags (q_1).** Increasing the AIGC-supply lag order raised F from 4.81 to 7.35. This pattern suggests that additional supply lags capture information about exposure allocation beyond the baseline one-day specification, strengthening evidence for a long-run relationship.

(3) **Weaker evidence under joint lag expansion.** Increasing all three lag orders (p, q_1, q_2) together yielded $F = 3.08$ at $(2, 2, 2)$, $F = 3.43$ at $(3, 3, 3)$, $F = 3.75$ at $(4, 4, 4)$ and $F = 3.61$ at $(5, 5, 5)$, all below the lower bound. Joint expansion introduces more short-run terms, increasing model complexity and reducing precision unless those terms contribute sufficient additional information. Here, the longer joint specifications weakened rather than strengthened evidence for the lagged-level relationship.

Across specifications supporting a long-run relationship, the estimated multiplier remained close to zero. The small estimated association is therefore consistent across supported specifications, while the strength of evidence for the long-run relationship depends on how the short-run dynamics are specified.

Table OT6. **Lag-order sensitivity for the highest-exposure-tier ARDL equation.** The outcome is the 31-day tier share for newly uploaded AIGC receiving >1,000 exposures. Lag order (p, q_1, q_2) refers to the outcome, AIGC supply and non-AIGC supply; BIC selects $(1, 1, 1)$. Under the stated Case 3 bounds criteria with two supply regressors, the 5% F bounds are 3.80 and 4.81 and the upper t bound is −3.53. Class is supported (shown as Long-run) if $F \geq 4.81$ and $t \leq -3.53$, inconclusive if $3.80 \leq F < 4.81$, and unsupported if $F < 3.80$; all displayed t-statistics are below −3.53. θ is the long-run multiplier on AIGC upload counts, scaled by 10^6, and is reported only for supported specifications; dashes indicate that the criteria do not establish a level relationship. The reported statistics and multiplier p-values use Newey–West HAC inference with a 7-day bandwidth.

Order	F	t	Class	θ	p
Baseline					
(1,1,1)	4.81	−5.64	Long-run	−0.041	0.069
AIGC supply (q_1)					
(1,2,1)	4.91	−5.40	Long-run	−0.036	0.107
(1,3,1)	6.14	−5.37	Long-run	−0.027	0.256
(1,4,1)	7.07	−5.55	Long-run	−0.027	0.267
(1,5,1)	7.35	−5.75	Long-run	−0.019	0.437
Outcome (p)					
(2,1,1)	4.48	−5.35	Inconclusive	—	—
(3,1,1)	5.15	−5.29	Long-run	−0.042	0.055
(4,1,1)	5.36	−5.09	Long-run	−0.043	0.052
(5,1,1)	5.67	−5.03	Long-run	−0.044	0.035
non-AIGC supply (q_2)					
(1,1,2)	3.35	−5.59	Unsupported	—	—
(1,1,3)	4.06	−5.79	Inconclusive	—	—
(1,1,4)	4.79	−6.16	Inconclusive	—	—
(1,1,5)	5.27	−6.43	Long-run	−0.044	0.041
Both supplies					
(1,2,2)	3.41	−5.38	Unsupported	—	—
(1,3,3)	4.15	−5.37	Inconclusive	—	—
(1,4,4)	4.93	−5.58	Long-run	−0.027	0.254
(1,5,5)	5.34	−5.83	Long-run	−0.021	0.389
All three					
(2,2,2)	3.08	−5.01	Unsupported	—	—
(3,3,3)	3.43	−4.72	Unsupported	—	—
(4,4,4)	3.75	−4.62	Unsupported	—	—
(5,5,5)	3.61	−4.34	Unsupported	—	—

O3 Recursive-residual Diagnostic Paths

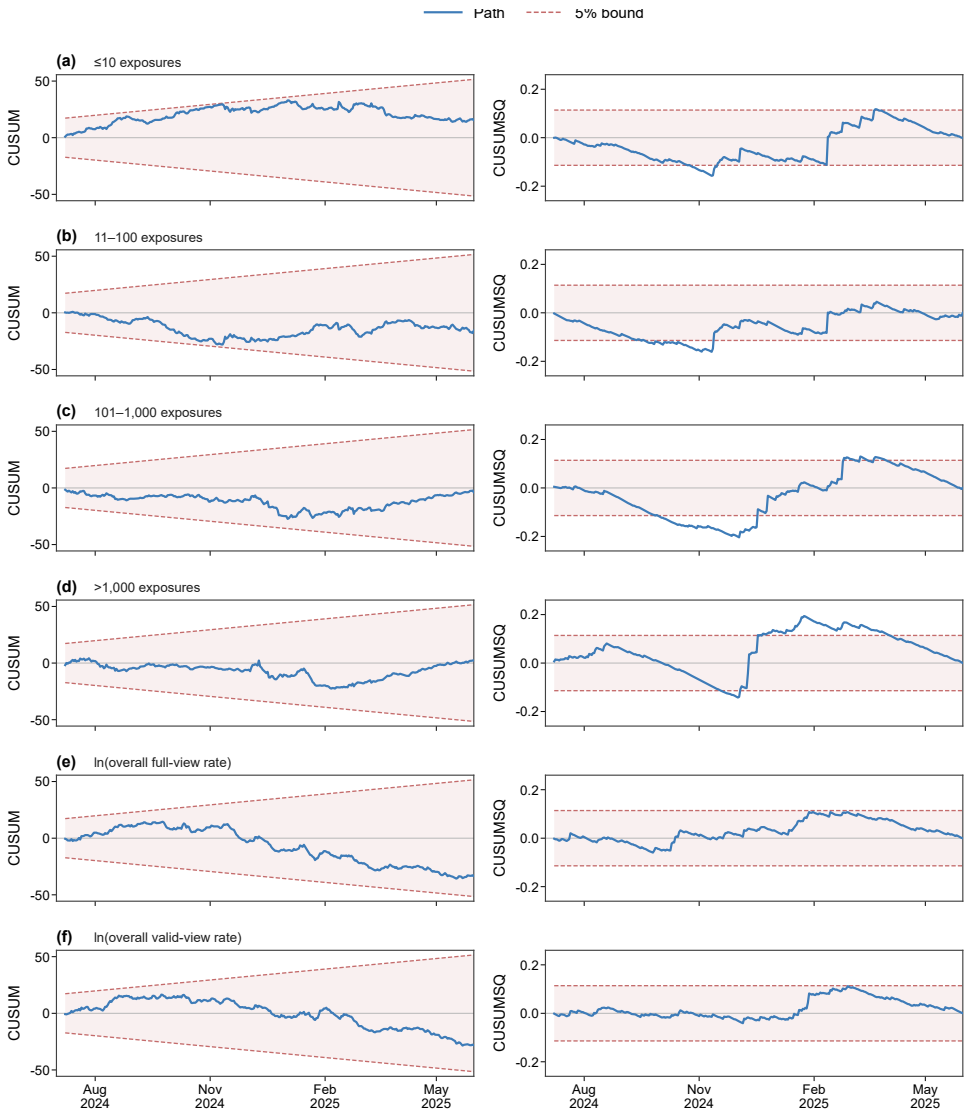


Fig. OF2. **Recursive-residual diagnostics for six ARDL equations.** Rows follow Table 10: ≤10 exposures (**a**); 11–100 exposures (**b**); 101–1,000 exposures (**c**); >1,000 exposures (**d**); ln(overall full-view rate) (**e**); and ln(overall valid-view rate) (**f**). Exposure-tier rows are the shares of newly uploaded AIGC videos receiving the specified numbers of exposures within 31 days. Each equation uses the BIC-selected order $(1, 1, 1)$. Left, cumulative sums of recursive residuals (CUSUM) with time-varying 5% bounds. Right, cumulative sums of squared recursive residuals (CUSUMSQ), centered by subtracting t/n, with flat 5% bounds of half-width 0.114. Blue lines show diagnostic paths; red dashed lines show bounds. All CUSUM paths remain within their bands. CUSUMSQ paths cross the bounds for the four exposure-tier equations but remain within them for the two engagement equations.

References

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